

Muhammad Akbar Khan



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RESEARCH PROFILE

MS Applied Mathematics researcher at NED University specializing in physics-informed neural networks and data-driven PDE solvers. First author or co-author of four research papers: one published in *Machine Learning: Science and Technology* (IOP Publishing) on an RFF–eikonal joint constraint framework for level-set interface advection; one under peer review at *Scientific Reports* (Springer Nature) on a Physics-Informed Neural Operator (PINO/FNO) for thermal ranking of low-cost wall materials in hot-dry climates; one paper under review at *Machine Learning: Science and Technology* on SDF-Aware Weighting with adaptive eikonal regularisation for three-dimensional level-set PINNs (MLST-105968); and one preprint on a Fourier Neural Operator for coupled hygrothermal ranking of wall materials under NASA POWER reanalysis forcing. Seeking a fully-funded PhD position to advance scientific machine learning at the intersection of numerical analysis and deep learning.

RESEARCH INTERESTS

Physics-Informed Neural Networks (PINNs) · Neural Operators · Scientific Machine Learning · Data-Driven PDE Modeling · Numerical Analysis · Level-Set Methods for Multiphase Flow · Computational Heat Transfer

EDUCATION

NED University of Engineering & Technology

Master of Science (MS) in Applied Mathematics

Aug 2024 – Dec 2026

Karachi, Pakistan

- CGPA: **3.44/4.00**
- Thesis: *A Systematic Study of Physics-Informed Neural Networks for the Level-Set Interface Advection*
- Supervisor: Dr. Fahim Raees (Associate Professor; PhD, TU Delft, 2016)
- Coursework:
 - *Core mathematics:* Differential Equations, Linear Algebra, Advanced Discrete Mathematics
 - *Computational methods & optimization:* Numerical Methods and Applications, Scientific Computing, Operations Research and Optimization
 - *Statistics & machine learning:* Applied Statistics, Statistical Method and Data Analysis, Fuzzy Logic and Neural Networks

International Islamic University Islamabad

Bachelor of Science (BS) in Mathematics

Jan 2017 – Jan 2021

Islamabad, Pakistan

- CGPA: **3.37/4.00**
- **HEC Need-Based Scholarship** (Higher Education Commission, Pakistan): full tuition and stipend for the duration of the BS program.

PUBLICATIONS

[1] **M. Akbar Khan** and F. Raees. *A Systematic Study of Physics-Informed Neural Networks for the Level-Set Interface Advection*. **Machine Learning: Science and Technology** (IOP Publishing), 2026. [doi:10.1088/2632-2153/ae8b74](https://doi.org/10.1088/2632-2153/ae8b74). Published July 2026. Gold Open Access, CC BY 4.0.

[2] **M. Akbar Khan**, F. Raees, and U. Fatima. *A Physics-Informed Neural Operator for Thermal*

Ranking of Low-Cost Wall Materials in Hot-Dry Climates. Scientific Reports (Springer Nature), 2026. Under peer review, July 2026.
[arXiv:2607.25668](https://arxiv.org/abs/2607.25668).

[3] M. Akbar Khan. *SDF-Aware Weighting: Adaptive Eikonal Regularisation for Three-Dimensional Level-Set Physics-Informed Neural Networks. Machine Learning: Science and Technology* (IOP Publishing), 2026. Under review; submitted August 2026 (MLST-105968).
[arXiv:2608.08322](https://arxiv.org/abs/2608.08322).

[4] F. Raees and M. Akbar Khan. *A Fourier Neural Operator for Coupled Hygrothermal Ranking of Low-Cost Wall Materials under NASA POWER Reanalysis Forcing*. Preprint, Zenodo, 2026.
[doi:10.5281/zenodo.21862964](https://doi.org/10.5281/zenodo.21862964).

RESEARCH EXPERIENCE

NED University of Engineering & Technology

March 2026 – Present

Research Assistant, Dept. of Mathematics

Karachi, Pakistan

- Appointed under **Sindh Research Project 321**: mathematical modeling for heat ventilation in rural housing.
- Developing finite-difference solvers and PINN models for heat transfer simulation in indigenous wall assemblies under Dr. Fahim Raees.

TEACHING EXPERIENCE

Cadet College Ormara

Aug 2022 – Dec 2025

Lecturer of Mathematics & OI/C FBISE Affairs

Gwadar, Pakistan

- Delivered mathematics lectures and tutorials at intermediate and higher-secondary level; designed and assessed examinations.
- Managed all Federal Board of Intermediate & Secondary Education (FBISE) academic affairs: exam registrations, compliance with board regulations, and direct correspondence with FBISE.

Excelsior School System

May 2021 – May 2022

Mathematics Teacher

Islamabad, Pakistan

- Planned and delivered mathematics lessons; set and marked examinations following the national curriculum.

TECHNICAL SKILLS

Scientific ML	PyTorch, Physics-Informed Neural Networks, Random Fourier Features, L-BFGS / Adam optimization
Programming	Python (NumPy, SciPy, Pandas, Matplotlib, Statsmodels, SQLAlchemy), SQL / PostgreSQL
Numerical Methods	Finite-difference and finite-volume schemes, classical ODE/IVP solvers, spectral methods, numerical linear algebra
Tools	Git, GitHub, VS Code, PyCharm Pro, Jupyter, L ^A T _E X (MiKTeX), TikZ
HPC	GPU training on Tesla T4, PyTorch 2.10, Python 3.12

OPEN-SOURCE PROJECT

PINN Level-Set Solver

[AkbarTheAnalyst/pinn-level-set](#)

PyTorch implementation underlying published MLST paper (doi:10.1088/2632-2153/ae8b74)

- Full PINN pipeline: 8-layer MLP with Tanh activations and Xavier initialization; RFF input embedding; causal weighting.
- Four canonical benchmarks (translation, rotation, reversed vortex, Zalesak disc) with automated ablation infrastructure.

CERTIFICATIONS

- AI for Everyone — DeepLearning.AI (Coursera)
- Google Data Analytics Professional Certificate — Google (Coursera)

LANGUAGES

English — Professional working proficiency · **Urdu** — Native

REFERENCES

Dr. Fahim Raees (MS Supervisor)

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Dr. Ubaida Fatima

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